

IMO Training – Number Theory

Miscellaneous Problems

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Notes

Unless otherwise specified, let p be a prime number.

Theorem. (Fermat's Little Theorem) $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$. Equivalently, $\forall a \in \mathbb{Z}$ with $(a, p) = 1, a^{p-1} \equiv 1 \pmod{p}$.

Definition. (Euler Totient Function) $\forall n \in \mathbb{N}, \varphi(n) = \#\{m \in \mathbb{N} \mid 1 \leq m < n, (m, n) = 1\}$. Equivalently, for $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ being distinct primes, and integers $a_1, a_2, \dots, a_k \geq 1$, $\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$.

Theorem. (Fermat-Euler Theorem) $\forall a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $(a, n) = 1, a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem. (Wilson's Theorem) p is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

Definition. (Order and Primitive Roots) For $a \in \mathbb{Z}, n \in \mathbb{N}$ with $(a, n) = 1$, the *order of a modulo n* ($\text{ord}_n(a)$) is defined by the minimal $m \in \mathbb{N}$ s.t. $a^m \equiv 1 \pmod{n}$. Furthermore, if $\text{ord}_n(a) = \varphi(n)$, then a is a *primitive root modulo n* .

Definition. (Valuation) For $m \in \mathbb{N}$, the *m -adic valuation* of n is the $a \in \mathbb{N}$ s.t. $m^a \mid n$ and $m^{a+1} \nmid n$.

Exercises

1. Let $n \in \mathbb{N}$. Find $(n! + 1, (n + 1)!)$.
2. (Iran 2005) Let $n, p > 1$ be positive integers and p be prime. It is given that $n \mid p - 1$ and $p \mid n^3 - 1$. Prove that $4p - 3$ is a perfect square.
3. Let $a, m, n \in \mathbb{N}$. Prove that $(a^m - 1, a^n - 1) = a^{(m,n)} - 1$.
4. Prove that for positive integers $a, b > 2$, $2^b - 1 \nmid 2^a + 1$.
5. If $a, b, p \in \mathbb{N}$ such that p is prime and $(a, b) = 1$, Prove that $\left(a + b, \frac{a^p + b^p}{a + b}\right) = 1$ or p .
6. (Japan 1996) Let $m, n \in \mathbb{Z}$ be odd and coprime. Find $(5^m + 7^m, 5^n + 7^n)$.
7. (St. Petersburg 1996) Find all $n \in \mathbb{N}$ such that $3^{n-1} + 5^{n-1} \mid 3^n + 5^n$.
8. (APMO 2002) Find all $a, b \in \mathbb{N}$ such that both $\frac{a^2 + b}{b^2 - a}$ and $\frac{b^2 + a}{a^2 - b}$ are integers.
9. (IMO 1998) Find all $x, y \in \mathbb{N}$ such that $xy^2 + y + 7 \mid x^2y + x + y$.
10. (IMO 1992) Find all $a, b, c \in \mathbb{N}$ with $1 < a < b < c$ such that $(a - 1)(b - 1)(c - 1) \mid abc - 1$.